SOOMIN HWANG

Frontend Developer (React / Next.js) · 4+ Years' Experience

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PROFILE

Frontend developer with a React/Next.js core, experienced across the full product lifecycle: backend API integration, CI/CD and test automation, and security review. Have built production systems including an AI-based protein structure analysis platform, a pathology image (WSI) analysis platform, and a smart-factory monitoring system. Applied AI development tools (Claude Code, Codex) to automate testing, code review, and refactoring workflows.

CORE SKILLS

Languages: JavaScript, TypeScript, Python

Frontend: React.js, Next.js (App Router), Vue.js/Vuetify, Zustand, Jotai, React Hook Form, TanStack Query (React Query), Tailwind CSS, MUI, Emotion, Vite, Three.js, Molstar, RDKit, OpenSeadragon, ApexCharts, Plotly

CI/CD & DevOps: Jenkins, Docker, Docker Compose, GitHub Actions, pnpm, Kubernetes (k3s), ArgoCD (GitOps)

Testing & Quality: Cypress, Playwright, MSW

Security: Snyk, OWASP ZAP, SonarQube

Tools & Platforms: GitHub, GitLab, Linux, Figma, Blender, Claude Code

PROFESSIONAL EXPERIENCE

Arontier Co., Ltd. — Frontend Developer, Platform Team

Mar 2025 – Oct 2026

AI-driven drug discovery platform company, founded 2017 — develops AD3, an end-to-end AI drug-discovery platform (ad3.io)
Assistant Researcher · Frontend · Team size 2–6 across projects · Resigned following business direction change and team wind-down

AD3 — AI Protein Structure Analysis Platform ***(ad3.io)***

- Enhanced and maintained the React frontend and supported the Django backend for AD3, an AI-based protein structure analysis platform.
- Customized 3D protein structure and chemical molecule rendering modules built on Molstar and RDKit, and implemented complex analysis-parameter input forms with React Hook Form, improving the validation structure.
- Led a code-splitting strategy using bundle analysis (rollup-plugin-visualizer) and manualChunks configuration, separating large libraries (MUI, Molstar, Ketcher, Plotly) into independent chunks.
- Diagnosed and resolved a rendering-delay issue in the RDKit-based Compound Designer (interaction lag beyond 150 nodes) through useEffect dependency optimization; applied lazy loading to react-plotly.js and verified the improvement via Lighthouse Total Blocking Time (TBT).
- Identified and fixed a layout-shift regression introduced by the lazy-loading rollout, and led the migration of the E2E test suite from Cypress to Playwright.
- Impact: reduced the main bundle from 32MB to 6.5MB (~80%) and improved the Lighthouse performance score from 29 to 78.

PATH — Pathology Image (WSI) Analysis Platform *(private — on-premise deployment)*

- Developed the viewer and interaction UI for a pipeline covering pathology slide upload through tiled image analysis, customizing an OpenSeadragon-based Whole Slide Image (WSI) viewer.
- Implemented state management with Jotai, forms with React Hook Form, and server-state/caching with TanStack Query.
- Addressed OWASP-aligned security requirements for an on-premise medical institution deployment, running OWASP ZAP scans and remediating findings.
- Impact: delivered the platform to an on-premise medical institution environment with no Critical or High severity vulnerabilities at delivery.

ADC — Antibody-Drug Conjugate Data Visualization ***(adc.arontier.co)***

- Built 3D protein and chemical-structure viewers on Molstar and RDKit using Next.js 16 (App Router) and Tailwind CSS.
- Introduced a Claude Code–based workflow to auto-generate Playwright tests, establishing the project's E2E testing framework.

Release Automation, E2E Testing & DevSecOps

- Designed and operated the shared CI/CD and security pipeline used across Arontier's frontend projects (AD3, PATH, ADC), including a Jenkins-based Build–Test–Scan–Deploy pipeline and a Docker-based Cypress E2E environment.
- Introduced Snyk and SonarQube for security and quality checks with automated reporting, and ran regular full-regression testing that caught side-effect regressions in out-of-scope features before release.
- Impact: cut manual regression testing time from roughly 5 hours to 1 hour (~80%).

Illuminarian LLC — Frontend Developer, Development Team

Jun 2024 – Feb 2025

Fintech startup backed by U.S. investment firm Solariant Capital
Team size 3 · Company ceased operations due to funding difficulties

- Reduced the Docker image size and deployment time by migrating the build to Next.js 14 Standalone output and pnpm.
- Built a job-posting and applicant-management service plus an admin console on Next.js 14, using React Hook Form and React Query for forms, validation, and server state.
- Introduced Mock Service Worker (MSW) to let frontend development and testing start before backend APIs were ready, and stabilized multilingual (next-translate) SSG page builds.
- Impact: Docker image size reduced from 1.3GB to 500MB (~62%) and deployment time from 15 minutes to under 3 minutes (~80%); parallel development via MSW mocking reduced schedule risk.

Yura Corporation — Web Developer, IT Division

Oct 2022 – May 2024

Global automotive electronics manufacturer (wiring harnesses, junction boxes) supplying Hyundai and Kia, with plants worldwide including Serbia
Senior Researcher · Team size 6 · Left due to company-wide reorganization

- Built a chart-based monitoring dashboard for a Serbian smart-factory production plant, owning the process end to end: Figma-based UI/UX design, React/MUI implementation, and Spring Boot API development.
- Implemented AOI (Automated Optical Inspection) first-pass-yield visualizations by product and production line using ApexCharts.
- Collaborated in English with the on-site team in Serbia to translate process formulas and feedback into the dashboard.
- Impact: real-time monitoring contributed to an approximately 20% reduction in AOI defect rate.

commON srl — Web Development Intern

Nov 2021 – May 2022

Milan-based startup — fashion magazine and event-marketing platform (common-mag.com)
Team size 1

- Contributed to the official website and a magazine purchase/display platform for a Milan-based startup, implementing responsive screens with Vue.js and Vuetify.
- Collaborated with a multinational team on fashion-magazine and event-marketing web services, and improved portfolio UX to help visitors arriving from social media convert into business contacts.

EDUCATION

Hankuk University of Foreign Studies (Yongin Campus)

2018 – 2023

B.Eng. in Computer & Electronic Systems Engineering; minor in Spanish Interpretation & Translation — GPA 3.54/4.5

CERTIFICATIONS & LANGUAGES

- Languages: Korean (Native) · English (Professional working proficiency)
- Engineer, Information Processing (정보처리기사) — Human Resources Development Service of Korea, 2026
- OPIc — Intermediate High (English), 2026
- Microsoft Certified: Azure Fundamentals (AZ-900), 2022

PORTFOLIO

soominlab.com — portfolio site (self-hosted, k3s + ArgoCD GitOps)
github.com/Lajancia — code repositories
lajancia.github.io — technical blog